Hazidien Ramadhan Utomo

PORTFOLIO

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2025

ABOUT ME

GIS SPECIALIST

HI I'M HAZIDIEN RAMADHAN UTOMO

I am a 23-year-old Urban and Regional Planning student at Institut Teknologi Sepuluh Nopember (ITS) with a passion for social interaction and exploring new opportunities. Known for my extroverted nature, strong critical thinking, collaborative spirit, and high self-esteem, I thrive in dynamic environments. I have strong skills in spatial analysis, remote sensing, and urban modeling, and I am proficient in tools such as ArcGIS, Google Earth Engine (GEE), and GeoSOS-FLUS. I use these tools to support land use modeling, urban planning strategies, and transportation analysis. My academic background is reinforced by hands-on experience in managing spatial data and conducting geospatial analysis to address urban challenges. I was also awarded 3rd place in a national-level GIS Analyst Infographic Competition organized by Universitas Pendidikan Indonesia, demonstrating my ability to merge technical expertise with effective visual communication.

EDUCATION

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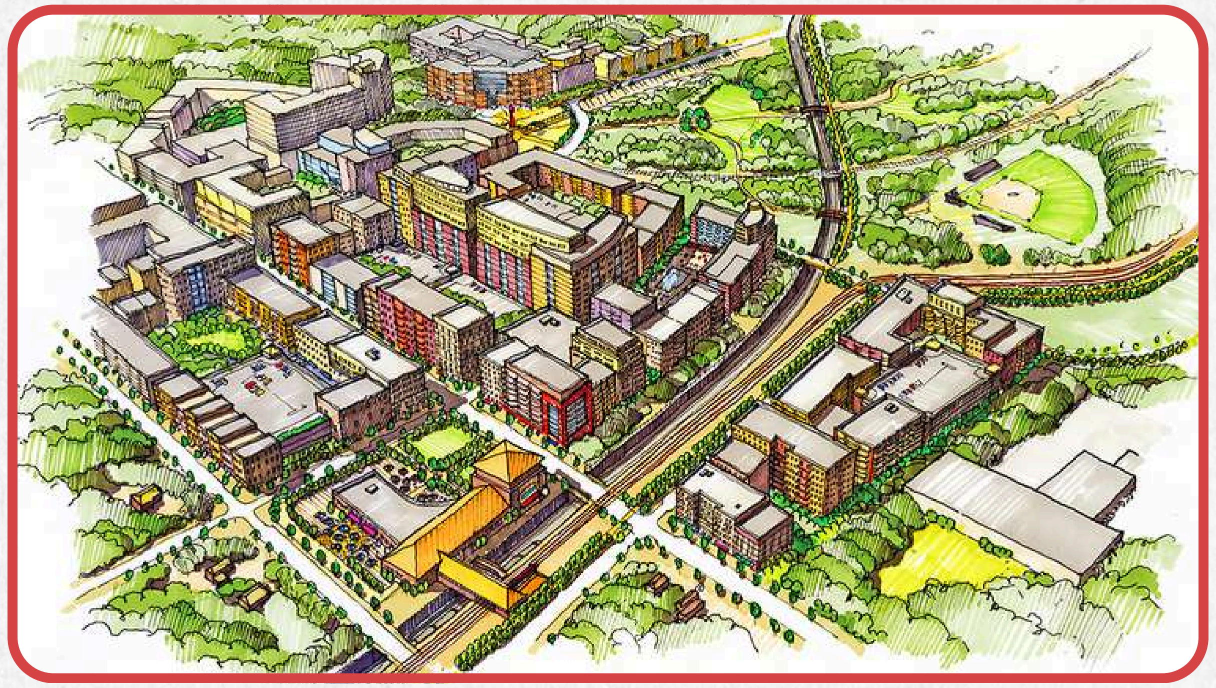
URBAN PLANNING

Urban and Regional Planning is a multidisciplinary field that focuses on designing and managing sustainable cities and regions. The program equips students with knowledge in spatial planning, land use, transportation, and urban design, supported by geospatial technology and data analysis.



- Urban Planning & Design Laboratory – Focuses on urban form, spatial planning, and urban design strategies.
- Regional Planning Laboratory – Explores regional development, policy, and spatial analysis at a wider territorial scale.
- Computation & Modeling Laboratory – Specializes in GIS, remote sensing, urban modeling, and transportation planning.

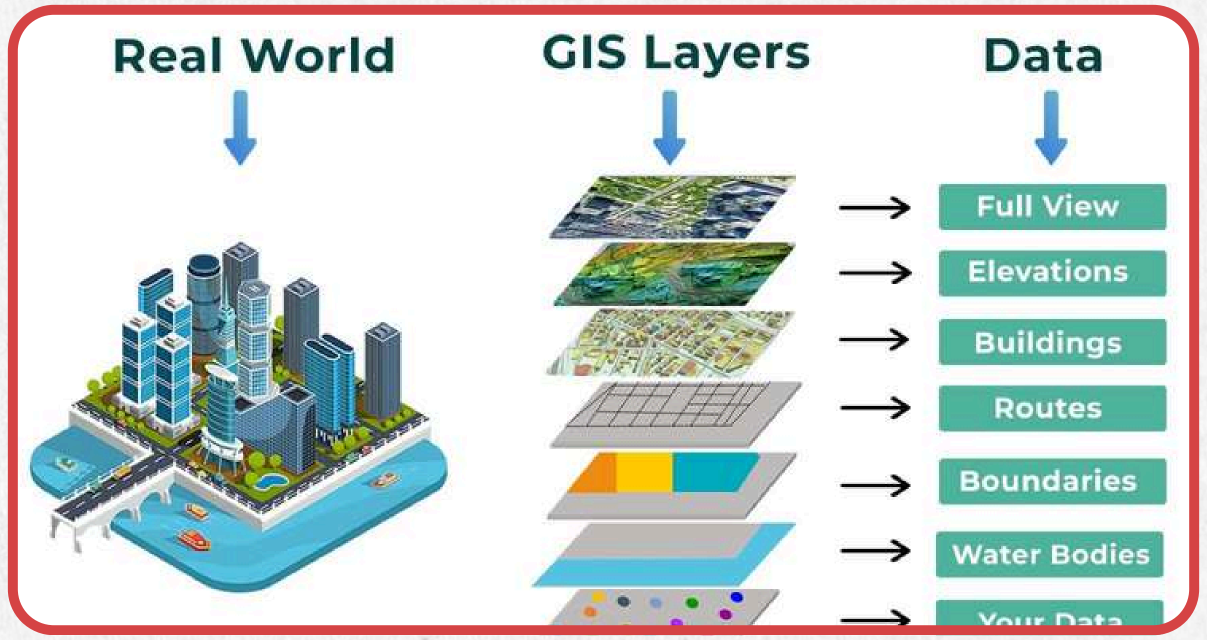
MY FOCUS

URBAN PLANNING STUDIO

Engaged in collaborative projects focusing on spatial planning, land use strategies, and urban design to create sustainable and livable cities.

GIS AND TRASNPORTATION ANALYST

Specialized in geospatial analysis, land use modeling, and transportation planning using tools such as ArcGIS, Google Earth Engine, and GeoSOS-FLUS.



SKILL

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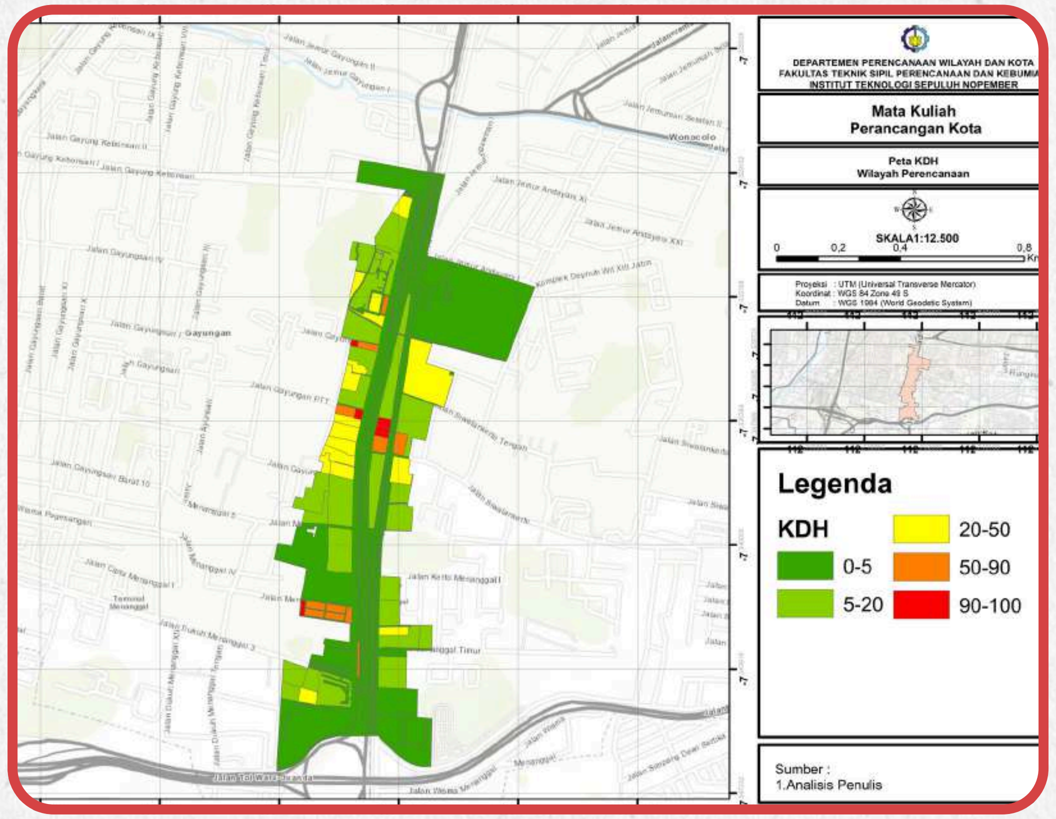
REMOTE SENSING

Conducting coastal and land cover analysis using Google Earth Engine (GEE) with indices such as NDTI, NDBI, NDWI, and others, as well as processing LiDAR data for spatial modeling and environmental monitoring.

GIS

Performing spatial data management and transportation analysis, including accessibility studies, land use modeling, and geospatial visualization using ArcGIS and other GIS tools.



URBAN DESIGN

Designing urban spaces and public areas with a focus on livability, walkability, and sustainability, integrating planning concepts into visual and practical design outputs.

WORK EXPERIENCE

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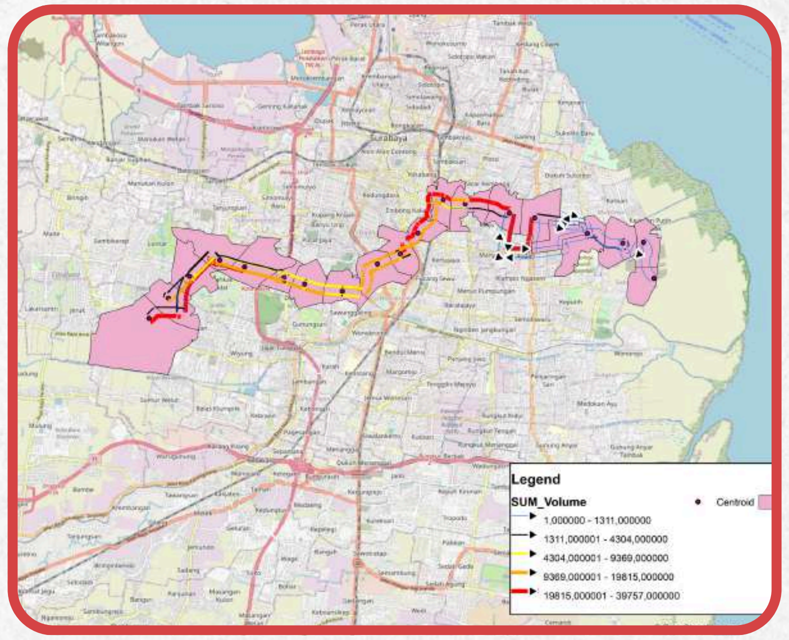
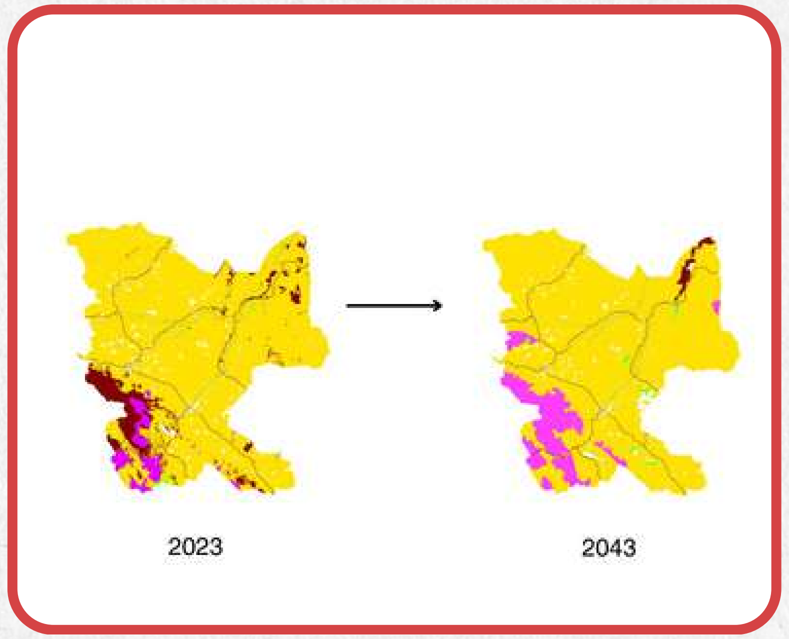
MRT SURVEYOR

Conducted household surveys and traffic counting to support urban mobility and transportation planning studies, including calculations using the Four Step Model (trip generation, trip distribution, mode choice, and route assignment).



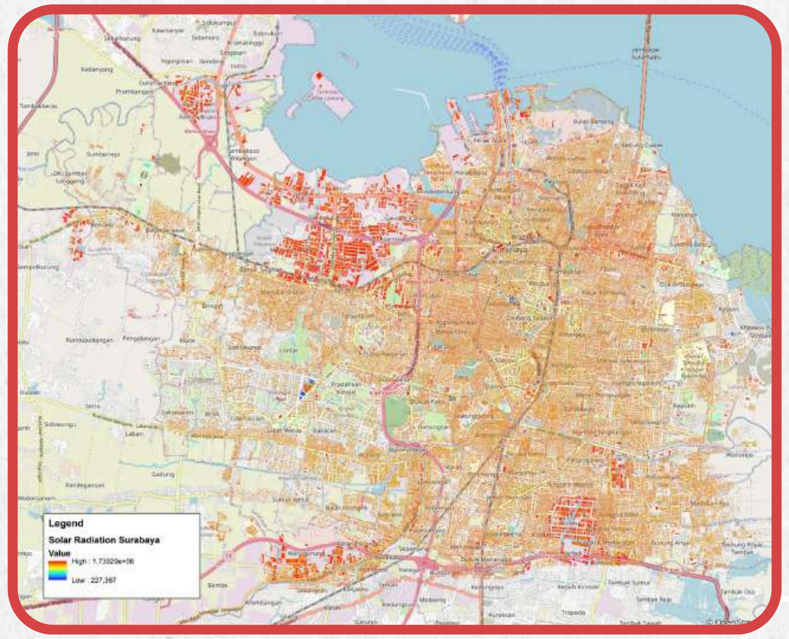
RESEARCH ANALYSIS

Performed land use change modeling using GeoSOS-FLUS in Rokan Hulu, focusing on spatial dynamics and future urban growth projection.



GREENOVA

Assisted in spatial analysis for solar energy potential modeling in Surabaya by processing LiDAR data into nDSM to identify suitable areas for solar panel installation.



PROJECT PORTFOLIO

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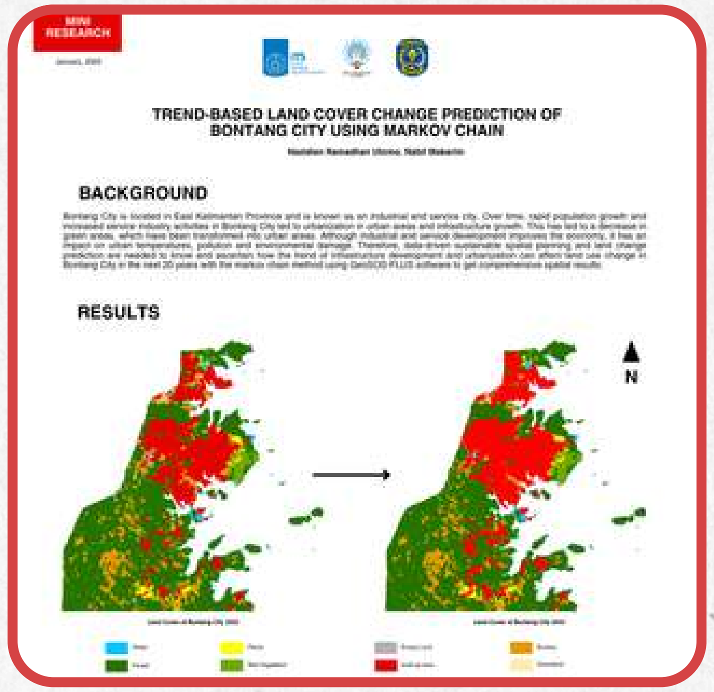

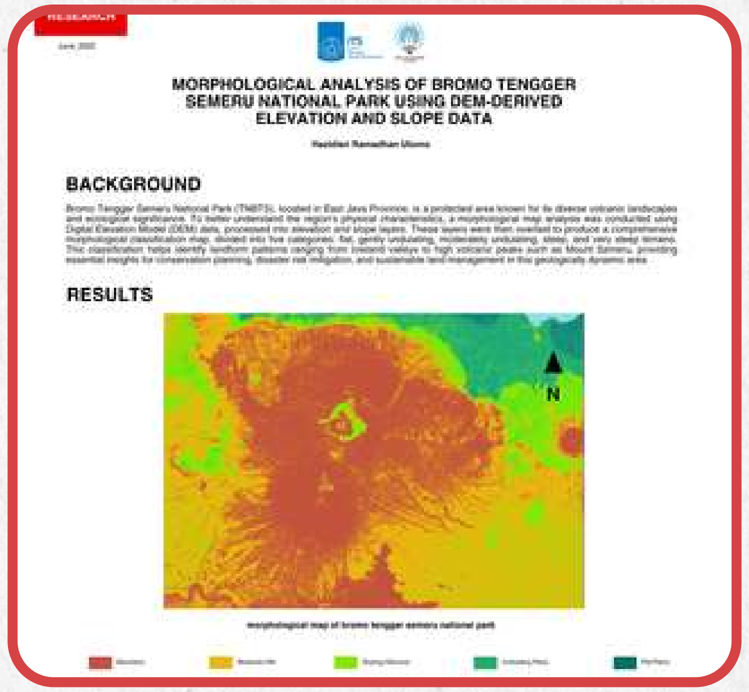

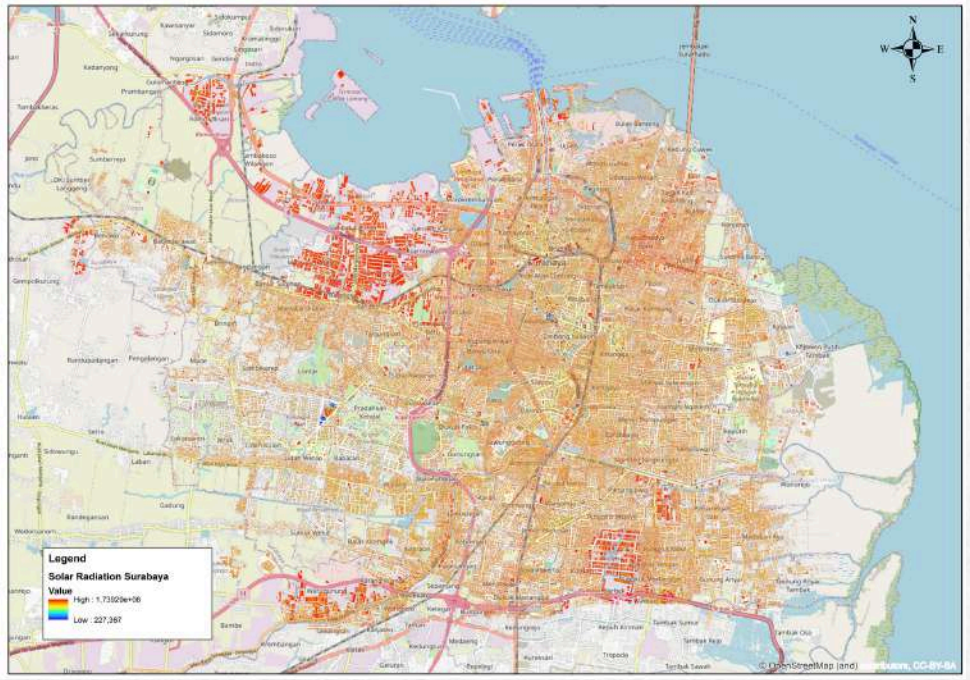

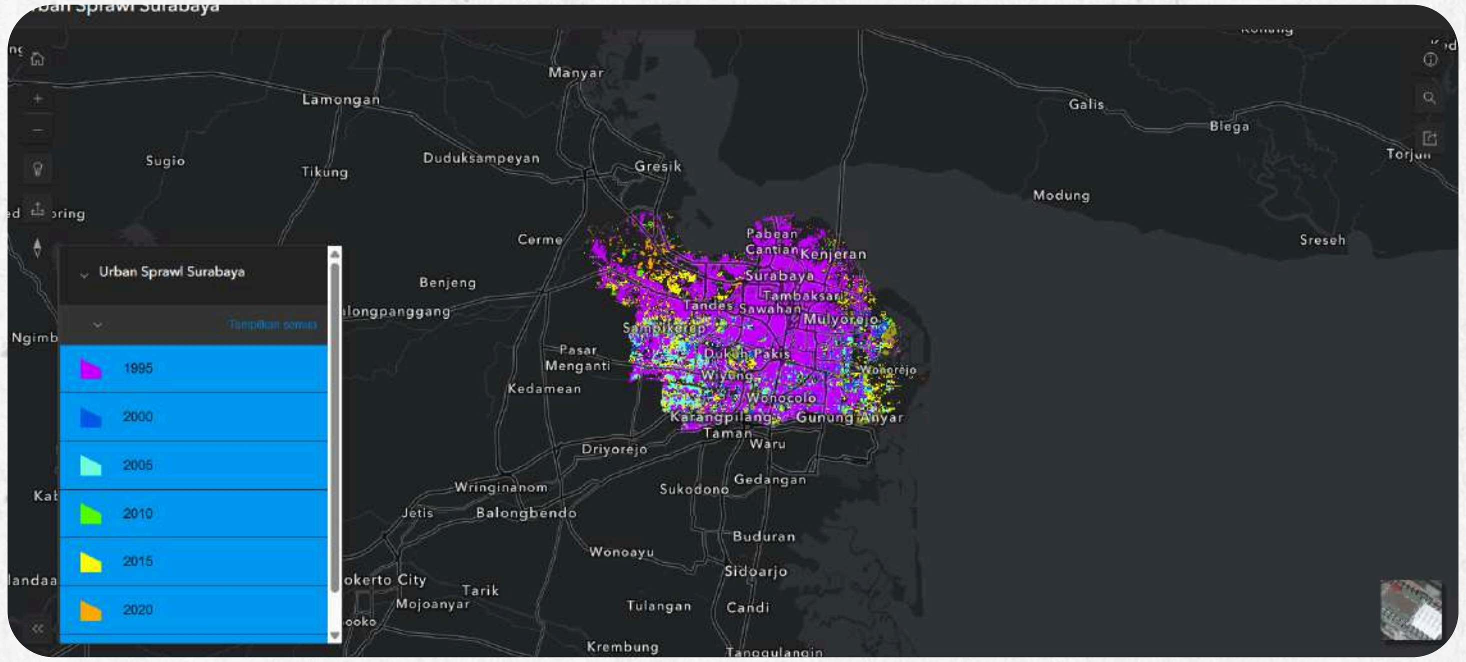

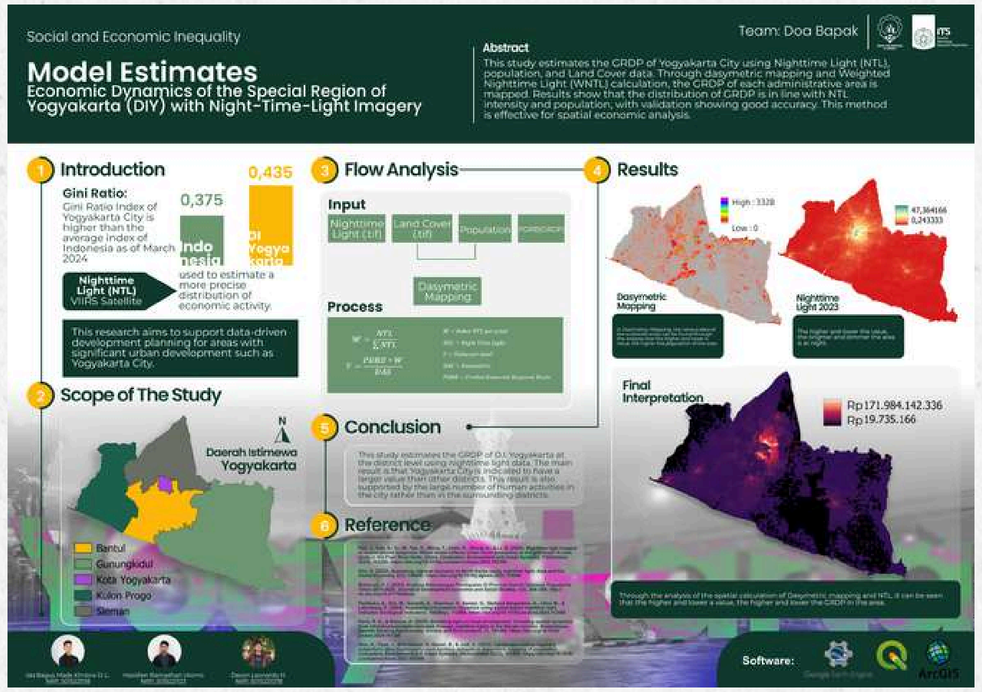

PROJECT 1

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LAND COVER CHANGES AT BONTANG CITY

This project predicts land cover change in Bontang City over the next 20 years using the Markov Chain method in GeoSOS-FLUS. Results show an 8.28 km² increase in built-up areas and significant loss of forests and vegetation due to urbanization and industrial growth. Recommendations include promoting vertical housing, preserving green areas, and improving waste management

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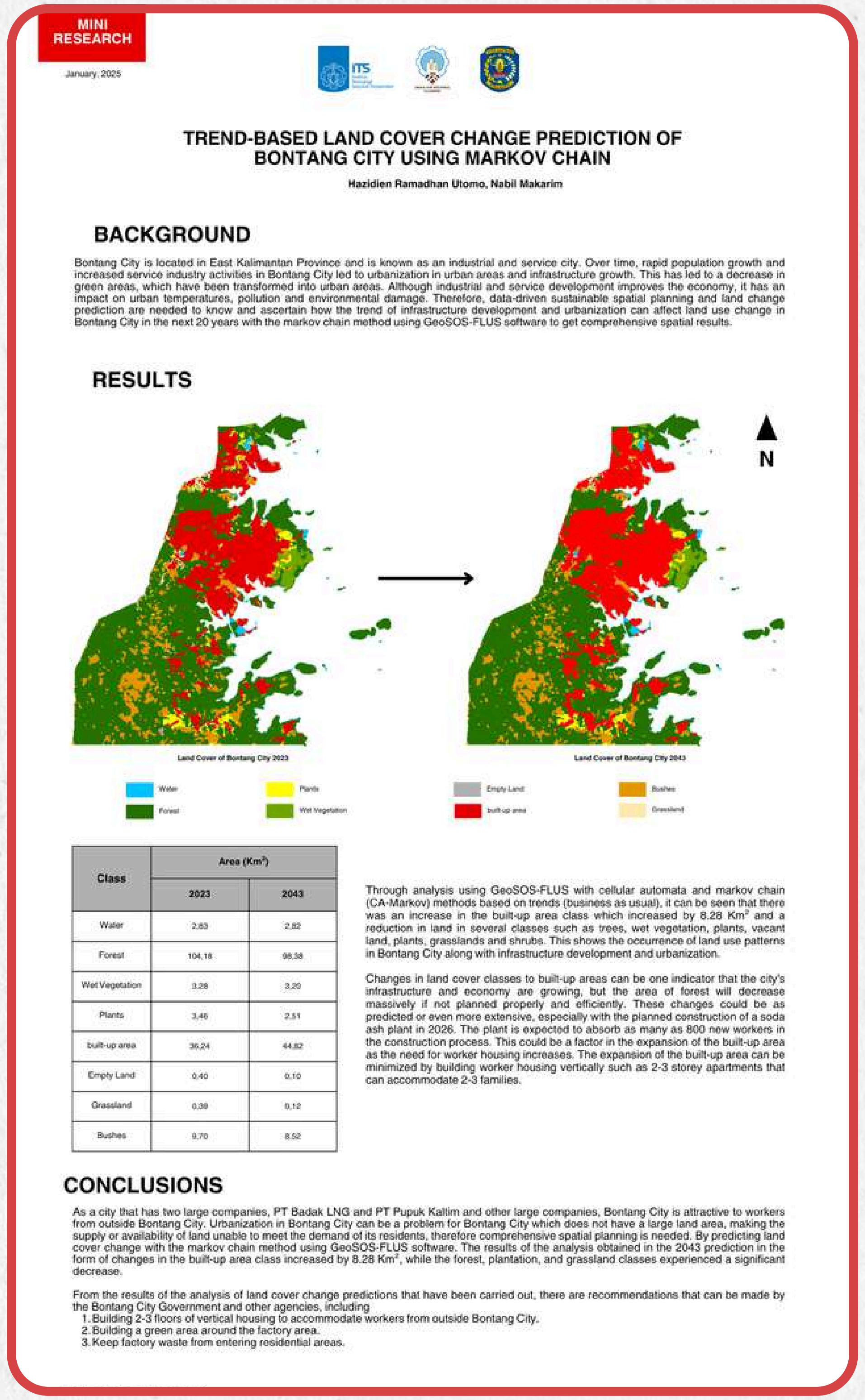

MINI RESEARCH

January, 2025

TRENDS-BASED LAND COVER CHANGE PREDICTION OF BONTANG CITY USING MARKOV CHAIN

Hazidien Ramadhan Utomo, Nabil Makarim

BACKGROUND

Bontang City is located in East Kalimantan Province and is known as an industrial and service city. Over time, rapid population growth and increased service industry activities in Bontang City led to urbanization in urban areas and infrastructure growth. This has led to a decrease in green areas, which have been transformed into urban areas. Although industrial and service development improves the economy, it has an impact on urban temperatures, pollution and environmental damage. Therefore, data-driven sustainable spatial planning and land change prediction are needed to know and ascertain how the trend of infrastructure development and urbanization can affect land use change in Bontang City in the next 20 years with the markov chain method using GeoSOS-FLUS software to get comprehensive spatial results.

RESULTS

Class	Area (Km^2)	
	2023	2043
Water	2,83	2,82
Forest	104,18	98,38
Wet Vegetation	3,28	3,20
Plants	3,46	2,51
built-up area	36,24	44,82
Empty Land	0,40	0,10
Grassland	0,39	0,12
Bushes	9,70	8,52

Through analysis using GeoSOS-FLUS with cellular automata and markov chain (CA-Markov) methods based on trends (business as usual), it can be seen that there was an increase in the built-up area class which increased by 8.28 Km^2 and a reduction in land in several classes such as trees, wet vegetation, plants, vacant land, plants, grasslands and shrubs. This shows the occurrence of land use patterns in Bontang City along with infrastructure development and urbanization.

Changes in land cover classes to built-up areas can be one indicator that the city's infrastructure and economy are growing, but the area of forest will decrease massively if not planned properly and efficiently. These changes could be as predicted or even more extensive, especially with the planned construction of a soda ash plant in 2026. The plant is expected to absorb as many as 800 new workers in the construction process. This could be a factor in the expansion of the built-up area as the need for worker housing increases. The expansion of the built-up area can be minimized by building worker housing vertically such as 2-3 storey apartments that can accommodate 2-3 families.

CONCLUSIONS

As a city that has two large companies, PT Badak LNG and PT Pupuk Kaltim and other large companies, Bontang City is attractive to workers from outside Bontang City. Urbanization in Bontang City can be a problem for Bontang City which does not have a large land area, making the supply or availability of land unable to meet the demand of its residents, therefore comprehensive spatial planning is needed. By predicting land cover change with the markov chain method using GeoSOS-FLUS software. The results of the analysis obtained in the 2043 prediction in the form of changes in the built-up area class increased by 8.28 Km^2, while the forest, plantation, and grassland classes experienced a significant decrease.

From the results of the analysis of land cover change predictions that have been carried out, there are recommendations that can be made by the Bontang City Government and other agencies, including

1. Building 2-3 floors of vertical housing to accommodate workers from outside Bontang City.
2. Building a green area around the factory area.
3. Keep factory waste from entering residential areas.

PROJECT 2

gis portfolio

MORPHOLOGICAL ANALYSIS OF BROMO TENGGER SEMERU NATIONAL PARK USING DEM-DERIVED ELEVATION AND SLOPE DATA

This study maps the morphology of Bromo Tengger Semeru National Park using DEM-derived elevation and slope data. Results show that over 75% of the area is dominated by steep and moderately steep terrain, shaped by volcanic activity around Mount Semeru and Bromo. These findings provide essential insights for conservation planning, disaster risk mitigation, and sustainable land management.

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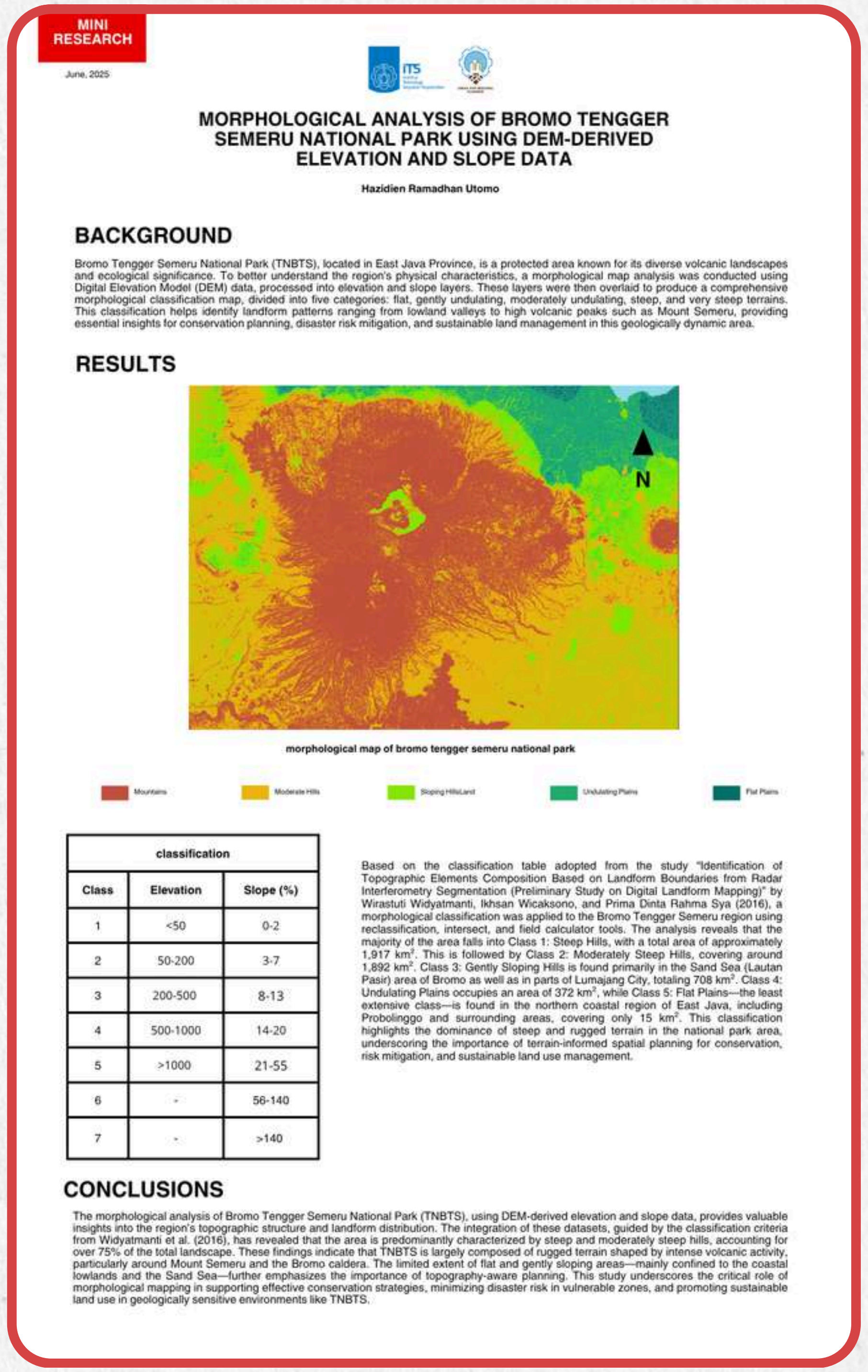

MINI RESEARCH

June, 2025

MORPHOLOGICAL ANALYSIS OF BROMO TENGGER SEMERU NATIONAL PARK USING DEM-DERIVED ELEVATION AND SLOPE DATA

Hazidien Ramadhan Utomo

BACKGROUND

Bromo Tengger Semeru National Park (TNBTS), located in East Java Province, is a protected area known for its diverse volcanic landscapes and ecological significance. To better understand the region's physical characteristics, a morphological map analysis was conducted using Digital Elevation Model (DEM) data, processed into elevation and slope layers. These layers were then overlaid to produce a comprehensive morphological classification map, divided into five categories: flat, gently undulating, moderately undulating, steep, and very steep terrains. This classification helps identify landform patterns ranging from lowland valleys to high volcanic peaks such as Mount Semeru, providing essential insights for conservation planning, disaster risk mitigation, and sustainable land management in this geologically dynamic area.

RESULTS

morphological map of bromo tengger semeru national park

Mountains · Moderate Hills · Sloping HillsLand · Undulating Plains · Flat Plains

classification		
Class	Elevation	Slope (%)
1	<50	0-2
2	50-200	3-7
3	200-500	8-13
4	500-1000	14-20
5	>1000	21-55
6	-	56-140
7	-	>140

Based on the classification table adopted from the study "Identification of Topographic Elements Composition Based on Landform Boundaries from Radar Interferometry Segmentation (Preliminary Study on Digital Landform Mapping)" by Wirastuti Widyatmanti, Ikhsan Wicaksono, and Prima Dinta Rahma Sya (2016), a morphological classification was applied to the Bromo Tengger Semeru region using reclassification, intersect, and field calculator tools. The analysis reveals that the majority of the area falls into Class 1: Steep Hills, with a total area of approximately 1,917 km^2. This is followed by Class 2: Moderately Steep Hills, covering around 1,892 km^2. Class 3: Gently Sloping Hills is found primarily in the Sand Sea (Lautan Pasir) area of Bromo as well as in parts of Lumajang City, totaling 708 km^2. Class 4: Undulating Plains occupies an area of 372 km^2, while Class 5: Flat Plains—the least extensive class—is found in the northern coastal region of East Java, including Probolinggo and surrounding areas, covering only 15 km^2. This classification highlights the dominance of steep and rugged terrain in the national park area, underscoring the importance of terrain-informed spatial planning for conservation, risk mitigation, and sustainable land use management.

CONCLUSIONS

The morphological analysis of Bromo Tengger Semeru National Park (TNBTS), using DEM-derived elevation and slope data, provides valuable insights into the region's topographic structure and landform distribution. The integration of these datasets, guided by the classification criteria from Widyatmanti et al. (2016), has revealed that the area is predominantly characterized by steep and moderately steep hills, accounting for over 75% of the total landscape. These findings indicate that TNBTS is largely composed of rugged terrain shaped by intense volcanic activity, particularly around Mount Semeru and the Bromo caldera. The limited extent of flat and gently sloping areas—mainly confined to the coastal lowlands and the Sand Sea—further emphasizes the importance of topography-aware planning. This study underscores the critical role of morphological mapping in supporting effective conservation strategies, minimizing disaster risk in vulnerable zones, and promoting sustainable land use in geologically sensitive environments like TNBTS.

PROJECT 3

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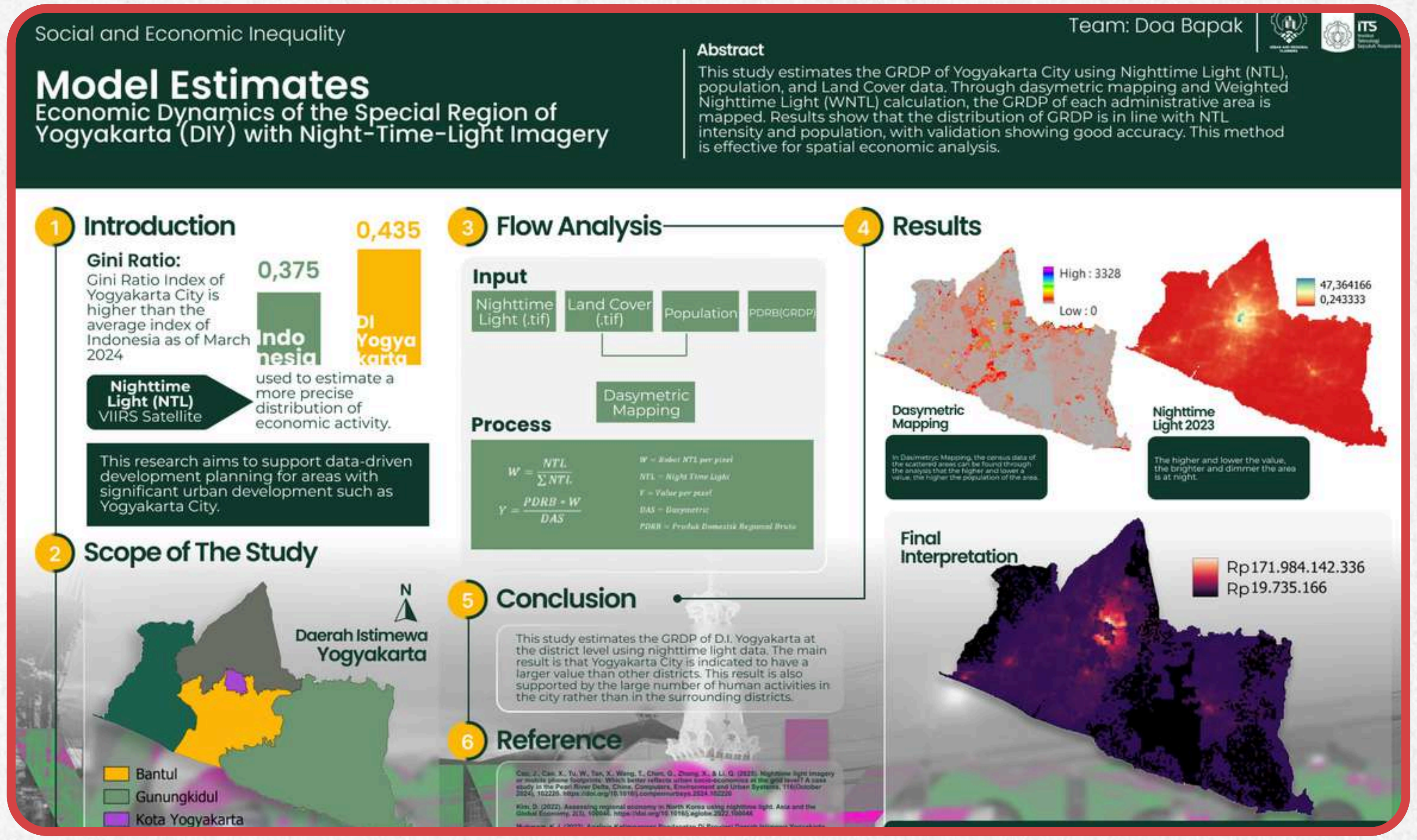

MODEL ESTIMATES

ECONOMIC DYNAMICS OF THE SPECIAL REGION OF YOGYAKARTA (DIY) WITH NIGHT-TIME-LIGHT IMAGERY

This study estimates the GRDP of Yogyakarta Special Region using Night-Time Light (NTL) imagery, land cover, and population data with dasymetric mapping. Results show that Yogyakarta City has the highest economic activity compared to other districts, supported by its high urban intensity. The findings demonstrate that NTL is effective for spatial economic analysis and supports data-driven development planning.

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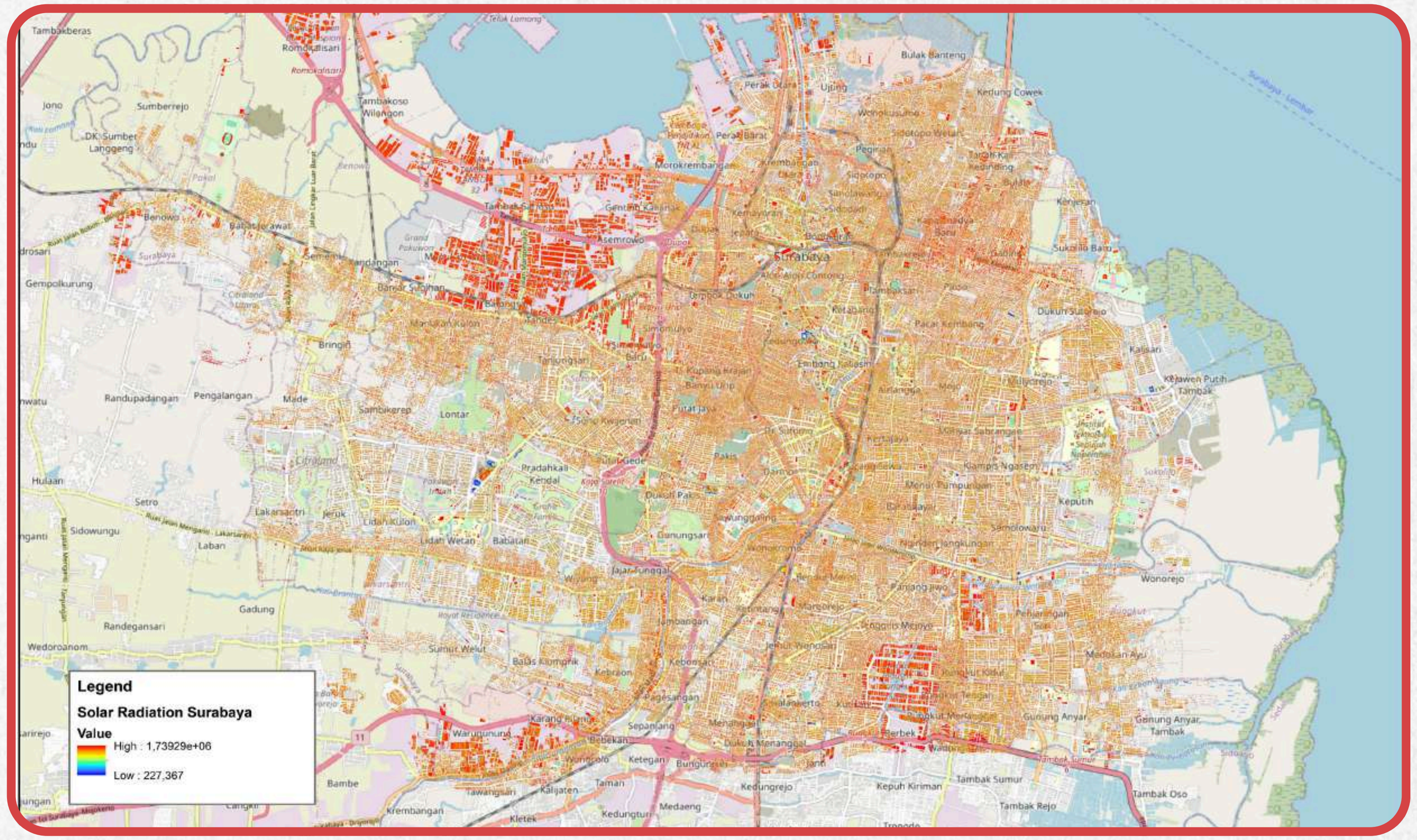
graphic designer

PROJECT 4

gis portfolio

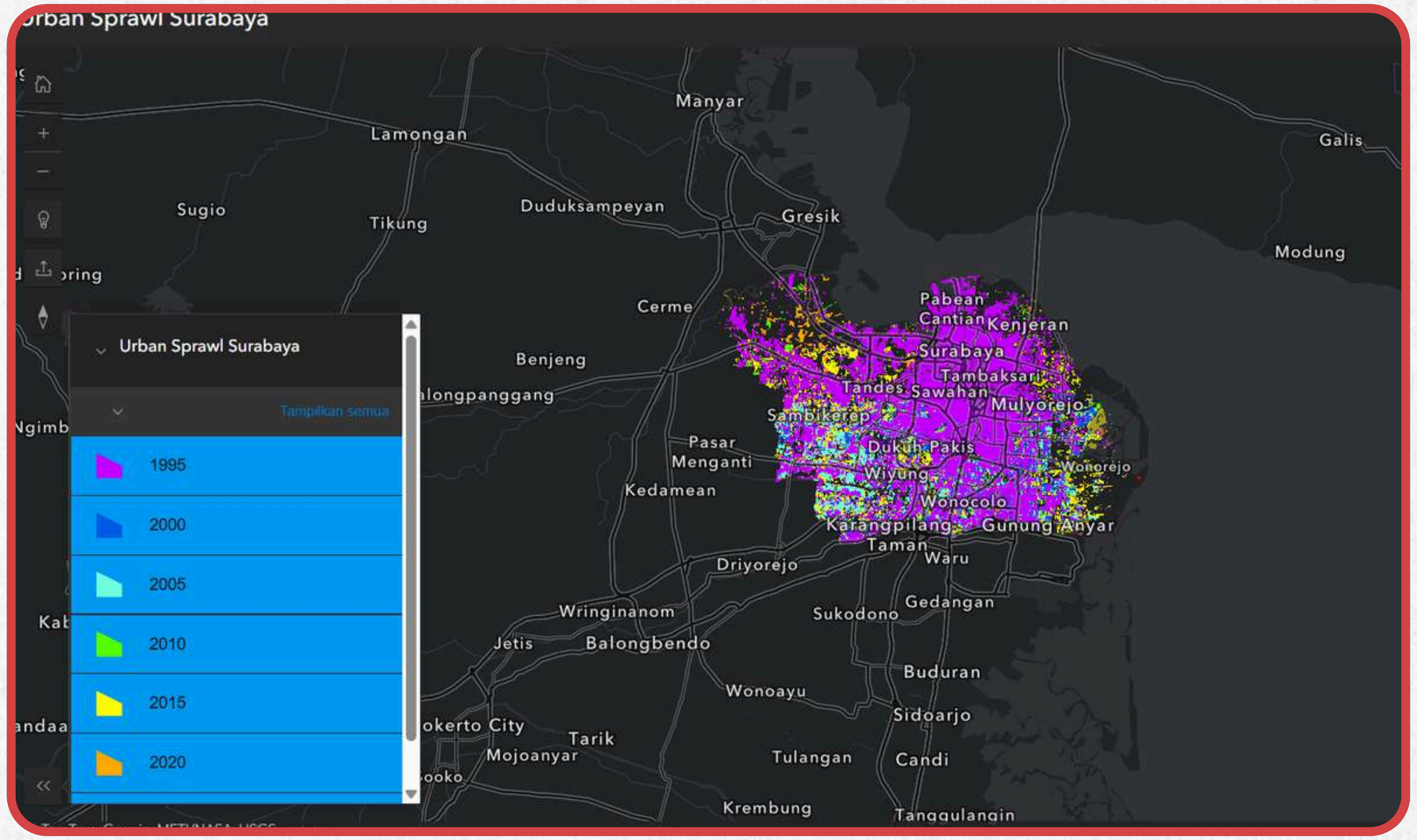
SOLAR ENERGY POTENTIAL ANALYSIS IN SURABAYA USING LIDAR-DERIVED NDSM AND ARCGIS SOLAR RADIATION TOOL



This project analyzes the solar energy potential of Surabaya using LiDAR-derived nDSM processed with the ArcGIS Area Solar Radiation tool, mapping annual rooftop solar potential and assessing spatial variability of solar irradiance (kWh/m²/year) by considering building orientation, shading, land cover, and urban density, and further estimating rooftop photovoltaic electricity generation potential of up to approximately 234 kWh/m²/year per panel based on solar radiation data, module efficiency, and performance ratio assumptions.

graphic designer

PROJECT 5



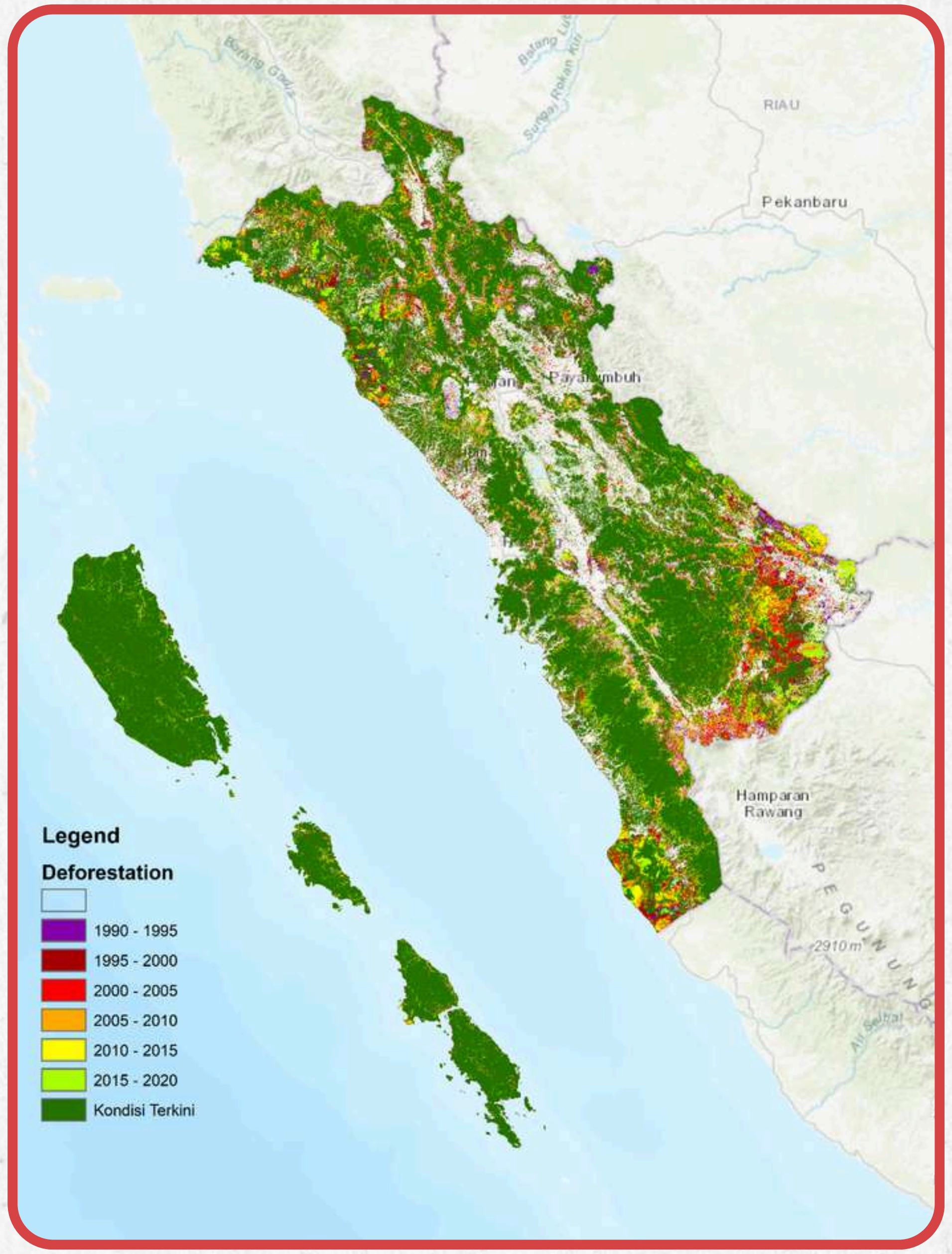
URBAN EXPANSION MAP AT SURABAYA CITY

This map shows the growth and distribution of population in Surabaya City from 1995 to 2025. The distribution of the population tends to point to the east and north. From this map, it is expected that stakeholders can make more appropriate policies in the future.

CLICK HERE TO ACCESS INTERACTIVE MAP

PROJECT 6

gis portfolio



SPATIO-TEMPORAL DEFORESTATION MODELING PROJECT – WEST SUMATRA

The Spatio-Temporal Deforestation Modeling Project in West Sumatra (Nov 2025–Dec 2025) involved analyzing multitemporal Landsat imagery from 2009 to 2024 using Google Earth Engine to identify patterns of forest loss, degradation, and regrowth, applying vegetation index–based thresholding and the Normalized Difference Fraction Index (NDFI) to detect deforestation dynamics, hotspots, and temporal change trajectories, and producing spatial outputs such as deforestation maps, hotspot analyses, and comparative method evaluations to support environmental monitoring and spatial planning decision-making.

CLICK HERE TO ACCESS PROJECT REPORT

LET'S WORK

TOGETHER!

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